

Nathaniel Mastracci

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EDUCATION

Sheridan College

Honours Bachelor of Computer Science (Mobile Computing)

Oakville, ON

Sep. 2019 – Dec 2023

Sheridan College

Computer Systems Technician - Software Engineering

Oakville, ON

Sep. 2015 – August 2017

EXPERIENCE

Freelance Web Developer

Mar. 2026 – Present

Freelance

Milton, ON

- Developed and deployed a website built using React, JavaScript, and Tailwind CSS
- Collaborated with client to gather requirements and deliver a custom and responsive website for their business
- Improved site performance and accessibility, enhancing user experience across devices

Produce Clerk

Jun. 2024 – Present

Metro

Milton, ON

- Maintained high-quality displays by rotating stock and removing damaged items, reducing waste
- Collaborated with team members to maintain an organized and visually appealing department
- Supported inventory accuracy by tracking stock levels and reporting shortages

Artificial Intelligence Research Assistant

Jan. 2023 – April 2024

Sheridan Centre for Applied AI

Oakville, ON

- Developed an AI model using Python and TensorFlow for the classification of heart sound audio captured through a electronic stethoscope
- Prepared a real-world dataset using preprocessing methods to improve model accuracy to 75%
- Published and presented a 6-page paper detailing our research efforts and development process

Artificial Intelligence Research Assistant

May 2022 – Aug. 2022

Sheridan Centre for Applied AI

Oakville, ON

- Developed cross-lingual voice clone models with Python and scikit-learn to produce AI-generated lip-synced videos
- Explored paid and open source voice cloning solutions to identify tools and features currently available
- Presented project outcomes to an industry partner

Frontend Developer

July 2019 – April 2020

Claret Asset Management

Montreal, QC

- Developed the frontend of a web application with Angular serving data from a REST API
- Created responsive UI features and implemented form validation and error handling to ensure reliable data input
- Collaborated on the development of a SQL database

PROJECTS

Collectors | TypeScript, Next.js, React, Tailwind CSS, Prisma ORM, Supabase, Vercel

Apr. 2026 – Present

- Developed and deployed a fullstack inventory management web application using Next.js, Supabase, and Vercel
- Integrated barcode scanning functionality to allow users to quickly add products by scanning UPC codes
- Connected an external product API to automatically retrieve item information from scanned barcodes
- Implemented OAuth authentication to support secure user sign-in and account management

Clearer | Kotlin, React, Firebase Firestore/Authentication, JUnit, Google Cloud Platform

Sep. 2022 – Dec. 2023

- Developed an assistive reading application using Kotlin and Firebase
- Created a web application in React to manage data stored in the backend
- Implemented OpenAI's API to create reading problems used in the application
- Utilized Google Cloud functions to collect user data

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C#, Kotlin, SQL (PostgreSQL, MySQL, MSSQL), HTML/CSS

Frontend: React, Angular, Tailwind CSS

Backend: Node.js, Prisma ORM, RESTful APIs

Tools & Platforms: Git, Postman, Vercel, Supabase, VS Code, Visual Studio, Google Cloud Platform